DS	**Session**	Tree 2	**Question Information**	Level 1 Challenge 80

Problem Description

Football is Monk's favourite sport, and his favourite team is "Manchester United." Manchester United has qualified for the Champions League Final, which will take place at London's Wembley Stadium. As a result, he decided to go watch his favourite team play.

When he arrived at the stadium, he noticed that there was a long wait for match tickets. He is aware that the stadium has M rows, each with a distinct seating capacity. They could or might not be comparable. The cost of a ticket is determined by the row. If there are K(always higher than 0) empty seats in a row, the ticket will cost K pounds (units of British Currency).

Now, every football fan standing in the line will get a ticket one by one.
Given the seating capacities of different rows, find the maximum possible pounds that the club will gain with the help of the ticket sales.

Constraints:
1 <= M <= 1000000
1 <= N <= 1000000
1 <= X[i] <= 1000000
Sum of X[i] for all 1 <= i <= M will always be greater than N.

Input:
The first line consists of M and N. M denotes the number of seating rows in the stadium and N denotes the number of football fans waiting in the line to get a ticket for the match.
Next line consists of M space separated integers X[1],X[2],X[3].... X[M] where X[i] denotes the number of empty seats initially in the ith row.

Output:
Print in a single line the maximum pounds the club will gain.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 4
1 2 4
```

EXPECTED OUTPUT

```
11
```

Test Case 2

INPUT (STDIN)

```
8 4
3 7 1 2 4 7 5 9
```

EXPECTED OUTPUT

```
31
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void heapify(int
arr[],int n,int i)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
6
```

Test Case 2

TOKEN COUNT

```
450
```

Test Case 3

NLOC

```
70
```

Code

monk_match_tickets.c

```c
#include <stdio.h>
#include <stdlib.h>

void heapify(int arr[],int n,int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;

    if (left < n && arr[left] > arr[largest])
        largest = left;
    if (right < n && arr[right] > arr[largest])
        largest = right;

    if (largest != i) {
        int temp = arr[i];
        arr[i] = arr[largest];
        arr[largest] = temp;
        heapify(arr, n, largest);
    }
}

int main() {
    int m, n;
    if (scanf("%d %d", &m, &n) != 2) return 0;

    int* heap = (int*)malloc(m * sizeof(int));
    for (int i = 0; i < m; i++) {
        scanf("%d", &heap[i]);
    }

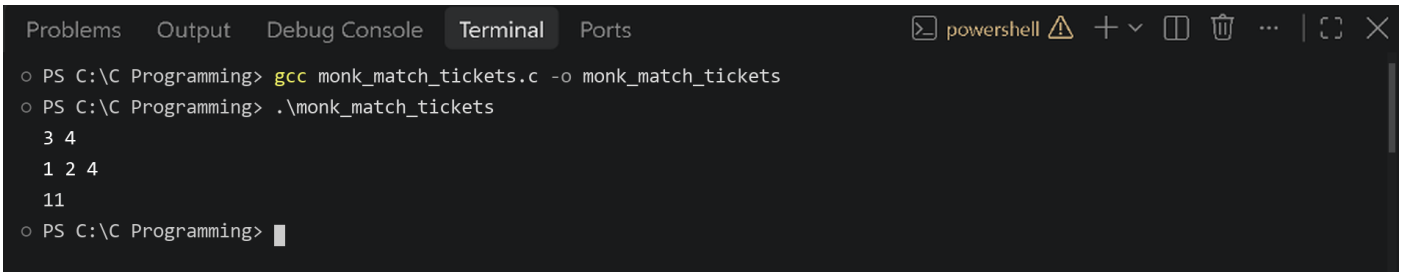
    for (int i = m / 2 - 1; i >= 0; i--) {
        heapify(heap, m, i);
    }

    long long total_revenue = 0;
    int heap_size = m;
    for (int i = 0; i < n; i++) {
        if (heap_size <= 0) break;
        int max_val = heap[0];
        total_revenue += max_val;

        if (max_val > 1) {
            heap[0] = max_val - 1;
            heapify(heap, heap_size, 0);
        } else {
            heap[0] = heap[heap_size - 1];
            heap_size--;
            if (heap_size > 0) {
                heapify(heap, heap_size, 0);
            }
        }
    }

    printf("%lld\n", total_revenue);
    free(heap);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc monk_match_tickets.c -o monk_match_tickets
PS C:\C Programming> .\monk_match_tickets
3 4
1 2 4
11
PS C:\C Programming>